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MILITARY SPECIFICATION

HOSE REEL ASSEMBLY, FUEL SERVICING, GENERAL REQUIREMENTS FOR

1. SCOPE

* 1.1 This specification covers the general requirements for the design and construction of a fuel servicing hose reel assembly that will accommodate standard diameter hoses, 1-1/2 through 4-inch size.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

Military

MIL-P-116	Preservation, Methods of
MIL-G-3056	Gasoline, Automotive, Combat
MIL-H-4441	Hose, Rubber, Gasoline, Lightweight
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-H-26521	Hose Assembly, Rubber, Fuel, Collapsible, Low Temperature with Reattachable Couplings

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage

FSC 4930

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MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-808	Finishes, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Fire Protection Association

Pamphlet 70 National Electrical Code (Electric Wiring and Apparatus)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

* 3.1 Detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable detail specifications. (See 6.2.)

3.2 Preproduction. This specification makes provisions for preproduction testing.

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3.3 Components. The hose reel assembly shall consist of the following major components:

Description	See Requirement
Drum assembly	3.8.1
Hub assembly	3.8.2
Motor and drive assembly	3.8.3
Frame	3.8.4.

3.4 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.5 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service, and entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished hose reel assembly.

3.5.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

* 3.5.2 Metals in contact with fuel. Unless otherwise specified (see 6.2), all metals in contact with the fuel shall be brass, bronze, or 300 series stainless steel.

3.5.3 Bonding. Metal parts shall be bonded together to prevent the possibility of sparking.

3.6 Design and construction. The hose reel assembly shall be of all-metal construction, self-contained, and designed and constructed to meet the performance requirements of this specification. The hose reel assembly shall withstand the stresses, impacts, vibrations, and other conditions incident to shipping, storage, installation, and service. The hose reel assembly shall be designed and constructed for maximum ease and safety of operation.

3.6.1 Reliability. The hose reel assembly shall have a minimum mean-time-between-failure (MTBF) of 100 hours with a minimum reliability of 0.9975 for an 0.25-hour mission at a confidence factor of 0.90. (See 6.3.1 through 6.3.3.

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(4) Level of preservation, packing, and marking if different (see 5.1.

3.6.2 Maintainability. The hose reel assembly shall be designed and constructed as specified herein and to comply with the following:

3.6.2.1 Accessibility. The design, construction, and component locations shall minimize the disturbance to components when adjustments, servicing, repairs, or other maintenance actions are made. All parts and components shall be located for ample and rapid access and clearance for maintenance actions with no disturbance of other components unless reliability or performance will be appreciably degraded or unless other requirements herein will be violated. If engineering study and analysis determines that a choice of the degree of accessibility must be made between two or more components, the position of each component shall be commensurate with the predicted frequency of maintenance on the component. That component expected to require the most frequent maintenance shall be most accessible. Positioning of any component providing poor or inconvenient accessibility when an obvious change could be made to improve accessibility and when there is no valid engineering reason why the change has not been made shall be grounds for rejecting the hose reel assembly.

3.6.2.2 Common components. The hose reel assembly shall be designed and constructed to provide for the maximum practicable interchangeability and minimum number of different parts and components including common hardware and fasteners such as bolts, screws, and nuts.

3.6.2.3 Parts. The equipment design shall be as simple as practicable and the number of parts minimized commensurate with the reliability, performance, and other requirements specified herein.

3.6.2.4 Fastening, retaining, and joining devices. Fastening, retaining, and joining devices or methods considered permanent or requiring destructive action to open, such as welded, riveted, or brazed joints or fastenings requiring peening or inelastic metal deformation, shall not be used unless necessary to assure compliance with the reliability, performance, or other requirements specified herein and unless the permanent device or method is common industry practice for similar applications. (An exception is that permanent methods of joining, fastening, or retaining may be used if analysis and experience with a similar application indicates the joint will not be opened or broken during the life of the equipment.)

3.6.2.5 Quick-disconnect fastenings. Covers, plates, or panels that must be removed for component adjustment, servicing, removal, or other maintenance shall be equipped with substantial, quick-disconnect fastenings.

3.6.2.6 Low temperature maintenance. To the maximum practicable extent, the equipment shall include maintainability provisions to facilitate maintenance

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and servicing and to allow ample clearance and accessibility for maintenance actions at low temperatures by personnel wearing heavy mittens and handicapped by bulky clothing and footgear.

3.6.2.7 Overtorque avoidance. Intricate locking devices, controls, and threaded fastenings that can be easily overtorqued and damaged by personnel lacking feel through thick gloves or due to numbness shall be avoided where possible.

3.6.2.8 Positioning for maintenance. Components expected to exhibit signs of excessive wear or other indication of impending or current failure shall be positioned, where practicable, to allow rapid inspection and determination that maintenance is needed.

3.6.2.9 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be so designed that the item can only be installed in its proper operating position.

3.6.2.10 Special tools. The hose reel assembly shall be so designed that special tools are not required for maintenance, assembly, and disassembly.

* 3.6.3 Safety. As a minimum, the contractor shall comply with the provisions of the paragraphs of MIL-STD-882 titled System safety criteria and considerations, Hazard levels, System safety precedence, and Design criteria/ specifications.

3.6.4 Functional design. The hose reel assembly shall be designed to satisfactorily service all fuels in accordance with MIL-G-3056, MIL-G-5572, and MIL-T-5624. The hose reel assembly shall be designed to accommodate collapsible hose in accordance with MIL-H-26521 and noncollapsible hose in accordance with MIL-H-4441. The hose reel assembly shall be designed to provide a servicing rate of from zero to the maximum flow rate required by the specification for the end item equipment at minimum pressure drop and shall allow evacuation of the noncollapsible hose. (See 6.2.)

3.6.5 General arrangement. The general arrangement of the hose reel assembly shall comply with the requirements specified herein and the requirements of the specification for the end item equipment. (See 6.2.) This specification establishes requirements for the following general types of hose reel assemblies:

a. Single-wrap. The single-wrap hose reel assembly shall incorporate a drum assembly with only sufficient width to allow one wrap of hose in contact with the drum surface. The succeeding wraps of hose shall wrap on the preceeding wrap of hose. The centerline of the hose shall essentially define a single plane for the initial and succeeding wraps for the complete length of hose.

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b. Multiwrap. The multiwrap hose reel assembly shall incorporate a drum assembly with sufficient width to allow two or more wraps of hose in contact with the drum surface.

3.6.6 Sizing for piping and joints. Hose reel assemblies to accommodate 1-1/2- and 2-inch-diameter hose shall incorporate inlet piping and swing joints of at least 2-inch ID. Hose reel assemblies to accommodate 2-1/2- and 3-inch diameter hose shall incorporate inlet piping and swing joints of at least 3-inch ID. Hose reel assemblies to accommodate 4-inch diameter hose shall incorporate inlet piping and swing joints of at least 4-inch ID.

3.6.7 Hose rollers. Hose rollers shall be provided to permit smooth and easy operation of the hose reel assembly and to prevent damage or excessive wear on the hose or any structure. Hose rollers shall be at least 1 inch in diameter and mounted in permanently lubricated bushings or other type bearings. The rollers shall be so installed that the hose cannot contact the roller mounting brackets. Two vertical rollers shall be installed to allow withdrawal of the hose at an angle of at least 30 degrees with a plane at right angles to the axis of the drum. A horizontal roller shall be provided at the bottom of the hose reel assembly or hose reel compartment. Additional rollers shall be provided as required when the angle of approach or departure with the surface or point of contact exceeds 10 degrees during withdrawal or recovery of the hose or where any edge or other structure could be subjected to or cause wear or damage to the hose. The approach and departure angle shall be determined during withdrawal with the end of the hose held at a length of 4 feet and during recovery of the hose with the hose laying on the ground. The hose rollers need not be installed as a part of the primary hose reel assembly.

3.6.8 Hose reels for noncollapsible hose. Any hose reel assembly to be utilized for noncollapsible hose shall be designed to service fuel at rated capacity in accordance with the specification for the end item equipment with any portion of the hose extended from the reel without causing any permanent deflection or distortion to the hose reel assembly.

* 3.6.9 Limit stop. The hose reel assembly shall be provided with an adjustable limit stop. The limit stop shall be adjustable such that one to zero wrap of hose remains on the drum for hose lengths of 60 to 100 percent of the hose storage capacity of the hose reel. The limit stop, once set, shall effectively and repeatedly stop the reel at the set position for a hose unwind speed of 5 fps as the hose approaches the set position and shall not cause any degradation or adverse effects to the hose reel.

* 3.6.9.1 Alternate limit stop. An alternate limit stop may be provided for hose reels for 1-1/2- and 2-inch ID noncollapsible hoses which will prevent

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the hose from being unwound beyond the normal fully extended position and avoid any bending and crimping of the hose. The limit stop shall be designed to effectively and repeatedly stop the reel at a hose unwind speed of 5 fps as the hose approaches the fully extended position without causing permanent distortion or damage to any part of the hose reel assembly.

* 3.6.10 Drag brake. The hose reel assembly shall be provided with an easily adjustable antibacklash drag brake. The brake shall be adjustable from zero to 100 pounds of additional force required to rotate the hose reel assembly in either direction. The drag brake shall be designed to absorb the frictional forces to be encountered during normal operation with the unwind forces and brake loads of 20, 35, and 50 pounds without excessive wear over long periods of normal operation.

3.6.11 Lubrication. Wherever practicable, lubricated-for-life components or components not requiring lubrication shall be used in the hose reel assembly design. Grease seals shall be designed and located to provide maximum accessibility for inspection and replacement and to prevent displacement, rupture, or other failure of seals due to excessive amounts of lubricant or lubricant pressure. Grease seals shall be so designed that pressure due to forced lubrication will not bind rotating parts. Lubricating grooves shall be provided as required to insure uniform lubrication of internal parts. Grease fittings shall be easily accessible.

3.7 Performance

3.7.1 Environmental conditions. The hose reel assembly shall be capable of satisfactorily operating under and withstanding the following environmental conditions:

3.7.1.1 Temperature ranges

3.7.1.1.1 Operating temperatures. The hose reel assembly shall be capable of performing as specified herein within ambient temperatures between -65° and +130°F.

3.7.1.1.2 Storage temperatures. The hose reel assembly shall not be damaged by storage in ambient temperatures from -80° to +160°F for not less than 50 hours.

3.7.1.2 Humidity. The hose reel assembly shall not be damaged by operation or storage in any relative humidity up to and including 100 percent, including conditions wherein condensation takes place in the form of water or frost.

3.7.1.3 Fungus. The hose reel assembly shall not be damaged by exposure to moist fungus growth such as encountered in tropical and subtropical climates.

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3.7.1.4 Salt atmosphere. The hose reel assembly shall not be damaged by operation or storage in an atmosphere containing salt-laden moisture as encountered near bodies of salt water and in transportation on shipboard.

3.7.1.5 Sand and dust. The hose reel assembly shall not be damaged by operation or storage in an atmosphere containing airborne sand and dust particles such as encountered in normal and desert operations.

3.7.2 Hydrostatic pressure and vacuum. The hose reel assembly shall be capable of withstanding a hydrostatic pressure of 0 to 200 psi and vacuum of 0 to 15 inches Hg without evidence of leakage, permanent deflection, or distortion.

3.7.3 Rewind rate. The hose reel shall provide an average rewind rate of 3 to 3.5 fps for the entire length of hose under normal operating conditions with the drag brake adjusted for a 20-pound force (see 3.6.10 and 6.2).

3.7.4 Rewind force. The hose reel drive mechanism shall provide a rewind force of at least 175 pounds without stalling or allowing slippage in the drive mechanism or damage to any portion of the hose reel assembly. The force shall be measured as a direct pull on the end of the hose when one-half of the hose length is extended from the reel.

* 3.7.5 Unwind force. The hose reel assembly shall require an unwinding force of no more than 15 pounds to maintain the unwind velocity. The force shall be measured as a direct pull on the hose when one-half of the hose length is extended from the reel.

3.8 Details of components

3.8.1 Drum assembly. The drum assembly shall consist of a spool, end discs, and supporting structure. The drum assembly shall be rigidly constructed to provide for long life under heavy service conditions. All exposed parts of the drum assembly shall be smoothly formed and free from sharp edges or corners which could cause wear or damage to the hose. The drum assembly shall be removable as a unit from the hose reel assembly frame.

* 3.8.1.1 Drum size. The drum assembly shall be of adequate size to accommodate the hose of the type, diameter, and length specified in the end item equipment specification (see 6.2) without extreme care in the winding operation, plus 25 percent additional capacity. When collapsible hose is used in conjunction with equipment incorporating a hose evacuation system, the size of the drum assembly shall be determined for the hose in the collapsible condition.

3.8.1.2 Drum spool. The drum spool shall be a minimum diameter of 10 inches for 1-1/2-inch hose and 14 inches for larger sizes as required to prevent

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crimping of the hose. The drum spool shall be a permanently formed cylinder constructed of sheet steel with a minimum thickness of 14 gauge or equivalent strength in other materials. The drum spool shall provide a continuous surface for wrapping of the hose with the exception of the area through which the hose connection extends, and this shall be held to a minimum. Exposed edges shall be rolled.

3.8.1.3 End discs. The end discs shall be smoothly formed sheet steel with a minimum thickness of 14 gauge or equivalent strength in other materials. The inner surfaces of the end discs shall be perpendicular to the axis of the drum. The end discs shall incorporate integrally formed reinforcement and structural spokes and concentric reinforcement as required to provide a rigid, long-life structure and shall comply with the loading requirements specified herein. All exposed edges shall be rolled. The outer edges of the end discs shall be rolled with a minimum roll diameter of 3/4 inch and through at least 180 degrees or 1/2 inch through 360 degrees. The end discs shall incorporate an integrally formed or other type grooved or raised section through 360 degrees to allow for alignment and positioning of the drum when assembled to the end discs. An alternate end disc design may be the use of concentric rings and spokes fabricated of suitable structural cross-section to provide a rigid, long-life structure and comply with the loading requirements specified herein. Spokes shall be positioned to provide for smooth recovery of the hose and shall be a maximum of 8 inches apart at the outer edge of the disc. The outer edge of the disc shall be constructed of at least 3/4-inch heavy steel tube if the disc is less than 3 feet in diameter and 1-inch heavy steel tube if the disc is more than 3 feet in diameter.

* 3.8.1.4 Supporting structure. The drum assembly shall be capable of assembly and disassembly. The end discs shall be assembled to the drum spool by means of bolts or threaded rods of 1/2-inch diameter or 3/8-inch diameter, as necessary. The bolts or rods with accompanying spacers, if required, shall support the surface of the drum spool. The bolts or rods shall be spaced no more than 6 inches apart. The supporting structure shall include other members as required in order to provide a permanently aligned and rigid drum assembly.

* 3.8.1.5 Structural loading requirements. The drum assembly as installed in the hose reel assembly shall be constructed to deflect no more than 3/8 inch when loaded by a 150-pound force parallel to the axis of the drum at any point on the outer edge of the end discs.

3.8.2 Hub assembly. The hub assembly shall consist of an inlet hub, rear hub, and interior piping.

3.8.2.1 Inlet hub. The inlet hub shall be fabricated of brass, bronze, or stainless-steel. The inlet hub shall provide rigid support and a free-running mounting for the drum assembly and shall incorporate a heavy-duty ball or roller bearing. The hose reel inlet may be either a pipe thread, flange, or groove-type connection (see 6.2).

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3.8.2.1.1 Swing joint. A heavy-duty, free-running, ball bearing swing joint shall be provided. The swing joint shall incorporate at least a dual O-ring or equivalent sealing design and shall be easily disassembled for seal replacement. The swing joint shall be fabricated of brass, bronze, or stainless-steel (see 6.2). The swing joint and inlet hub bearing shall incorporate a flanged connection incorporating an O-ring seal to mate with the interior piping. The swing joint and inlet hub bearing shall be detachable from the drum assembly without disassembling the drum assembly. The mating flange on the interior piping shall be tapped for the flange assembly bolts. The swing joint and inlet hub bearing may be an integral design.

3.8.2.2 Rear hub. The rear hub shall provide rigid support and a free running mounting for the drum assembly. The rear hub shall incorporate a heavy-duty, self-aligning ball or roller bearing.

3.8.2.3 Interior piping. The interior piping shall be fabricated of brass, bronze, or stainless-steel (see 6.2). The interior piping shall be designed to provide minimum pressure drop from the swing joint to the hose connection. The interior piping shall terminate outside of the drum and permit connection of the hose essentially tangential to the drum. Any outlet connection which results in crimping of the hose will be unacceptable. The portion of the interior piping external to the drum shall allow smooth and uniform consecutive layers of hose over the hose connection. The interior piping shall be so designed that it can be removed as a unit.

3.8.2.3.1 Hose connection. The hose connection shall incorporate a female threaded adapter. The female thread shall mate with the male threaded connection on the hose as required by the specification for the end item equipment (see 6.2). The hose adapter shall be detachable from the interior piping by means of a flanged joint incorporating an O-ring seal. The mating flange on the interior piping shall be tapped for the flange assembly bolts. The flange and flange assembly bolts shall be located to permit easy disassembly of the flange as a means of removing the hose.

3.8.2.4 Axial structure. Heavy-duty axial structure shall be provided between the rear hub and the interior piping in order to provide a continuous load-bearing member along the axis of the reel from one end disc to the other. The axial structure may be incorporated as an integral part of the interior piping. The interior piping and axial structure shall be so designed that they can be installed without the drum assembly, and a hose can be attached to the hose connection for normal servicing with the hose in a permanently extended condition.

* 3.8.3 Motor and drive assembly. The motor and drive assembly shall consist of an electric rewind motor, clutch (if necessary, see 3.8.3.2), gears or chain and sprockets, and associated electrical wiring and controls.

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* 3.8.3.1 Motor. The motor shall conform to NFPA pamphlet 70 for class 1, division 2, group D hazardous locations, and the contractor shall submit the label or listing of Underwriters' Laboratories, Incorporated as evidence of conforming to the explosionproof requirements. (See 6.4 and 6.4.1.) The motor shall be sized to comply with the rewind power requirements specified herein. The motor shall rewind the hose reel 10 consecutive times in as rapid succession as possible and then operate indefinitely to rewind the hose in 1-minute intervals. The motor shall accomplish the above when the drag brake is set for a 35-pound rewind force, and the motor shall not overheat. Unless otherwise specified (see 6.2), the motor shall be 12-V dc for use in an automotive electrical system.

* 3.8.3.2 Clutch. If necessary to meet the unwind force requirement, a clutch in accordance with the following shall be provided. The motor shall drive the reel through the clutch. The clutch shall be completely enclosed and shall provide for automatic engagement of the motor with the reel when the motor switch is actuated. The motor shall be disengaged at all other times in order that the hose may be extended without turning the motor shaft. The clutch shall provide for friction engagement or positive smooth mechanical engagement. A friction-type clutch shall be designed to lockup to an extent necessary to insure slippage and overheating will not occur under any operating condition. Any mechanical engagement device which can cause clashing of mechanical parts or heavy shock loading will be unsatisfactory. The clutch shall be designed to provide long life under field operating conditions. The clutch and motor shall be rigidly mounted and easy to remove from the hose reel either individually or as a unit without disassembling any other portion of the reel. Motor and clutch mountings shall be adjustable as required to allow for wear of other parts.

* 3.8.3.3 Gears or chain and sprockets. The hose reel assembly shall be entire gear driven or chain and sprocket driven. The gears, bearings, and chain and sprockets shall be designed for 1-1/2 times the load imposed during the 175-pound rewind force. The gears or chain and sprockets as applicable shall have a smooth, quiet operation and shall retain their positions once properly aligned without requiring periodic maintenance or tightening of components. The chain and sprockets shall have provisions for easy adjustment of the chain tension.

3.8.3.4 Electric wiring and controls. The electrical wiring installation and controls shall conform to NFPA pamphlet 70 for class 1, division 2, group D, hazardous locations. (See 6.4.) The contractor shall submit the label or listing of the Underwriters' Laboratories, Incorporated as evidence of conforming to the explosiveproof requirements for switches and other electrical devices.

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* 3.8.3.4.1 Control switch. The hose reel assembly shall be controlled by a deadman-type control switch. The switch shall be mounted for the convenience of the operator and, if appropriate, on the control panel of the end item equipment.

* 3.8.3.4.2 Overload protection. The hose reel assembly shall incorporate a manual reset circuit breaker. The circuit breaker shall prevent damage and overheating of the motor or other electrical components due to overload or malfunction. The circuit breaker shall be sized to provide the current sufficient to meet the power requirements specified herein without breaking. The circuit breaker shall be easily accessible and mounted in an area protected from the weather, external of the pumping compartment and hazardous vapor area (see 6.2). Under these conditions, the circuit breakers need not be explosionproof.

3.8.4 Frame. The hose reel assembly shall incorporate a single unit frame on which the other components are mounted. The frame shall be constructed of steel structural sections either bolted or welded to provide a completely rigid mounting for the drum and hub assembly and the motor and drive assemblies. The frame shall withstand the loading requirements specified herein. The frame shall be cross-braced for added strength. The frame shall be designed to provide adequate clearances for all rotating parts considering the wear and tear which will occur during the life of the hose reel assembly.

3.8.4.1 Structural loading requirements. The frame shall be constructed to deflect no more than 1/16 inch when loaded by a 750-pound force in any direction at either hub assembly mounting point. The frame shall be so constructed that there is no visible deflection when the motor is actuated to operate the reel.

3.8.4.2 Dual reel installation. If applicable, two reels may be mounted in a common frame. The reels shall incorporate separate motor and drive assemblies and shall be capable of completely independent operation. The frame shall incorporate a center mounting structure equivalent in strength to the outside end structure to provide mounting points for the rear hubs.

* 3.9 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

* 3.10 Electromagnetic interference. The hose reel assembly shall conform to the electromagnetic interference requirements of MIL-STD-461.

3.11 Screw threads. The screw threads in the hose reel assembly shall be in accordance with Handbook H28.

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3.12 Finishes and protective coatings

3.12.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808. A listing of the procedures used shall be furnished the procuring activity and shall be submitted with the report specified in 4.4.2.

3.12.2 Painting. Exposed metal surfaces shall be full-gloss yellow, color code DG, in accordance with MIL-STD-808. (See 6.2.)

3.12.3 Anodizing. All aluminum-alloy moving parts shall be given an anodic coating.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Workmanship. The hose reel assembly, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts; and tightness of assembly screws, and bolts, et cetera.

3.14.1 Gear and lever systems. Gear and lever systems shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots, and other irregularities.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the hose reel assembly shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

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4.3 Test conditions

4.3.1 Lubricants. Lubricants used in the hose reel bearings during the tests specified under 4.4.3 shall be as recommended by the manufacturer of the hose reel and shall conform to existing Government specifications used in this type service.

4.3.2 Test fluid. The test fluid shall be grade JP-4 turbine fuel in accordance with MIL-T-5624.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One hose reel assembly shall be subjected to the preproduction tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction tests, a preproduction test report in accordance with MIL-STD-831 shall be prepared and three complete copies of the report furnished to the procuring activity.

4.4.2.1 Explosionproof components. A copy of the Underwriters' Laboratories, Incorporated label or listing for the motor and other electrical devices shall be included as a part of an attachment to the preproduction test report.

4.4.2.2 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the above report (same quantity and quality of copies) or shall be included as part of the report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on use conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

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f. Estimates of manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests described under 4.6, conducted as follows:

a. Examination of product 4.6.1

b. Mechanical operation 4.6.2

c. Hydrostatic pressure and vacuum test 4.6.3

d. Operating hydrostatic pressure test 4.6.3.1

e. Unwind force test 4.6.4

f. Drag brake test 4.6.5

g. Rewind rate test 4.6.6

h. Rewind force test 4.6.7

i. Structural loading (drum assembly) test 4.6.8

j. Structural loading (frame) test 4.6.9

k. Endurance test 4.6.10

l. Electromagnetic interference test 4.6.11

m. Environmental tests 4.6.12

n. Reliability demonstration and test . . . . . 4.6.13.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each hose reel assembly shall be subjected to the following tests as described under 4.6:

a. Examination of product 4.6.1

b. Mechanical operation 4.6.2

c. Operating hydrostatic pressure test 4.6.3.1

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d. Unwind force test 4.6.4

e. Rewind rate test 4.6.6

f. Rewind force test . . . . . 4.6.7.

4.6 Test methods. All tests shall be conducted using the identical hose required by the specification for the end item equipment.

4.6.1 Examination of product. The hose reel assembly shall be inspected to determine compliance with the requirements specified herein and the applicable requirements of the specification for the end item equipment. Particular attention shall be given to materials, workmanship, dimensions, protective finishes and their application, welding, riveting, and markings.

4.6.2 Mechanical operation. A thorough check shall be made of the hose reel assembly to insure the proper functioning of all mechanical parts, interchangeability of removable parts, installation of equipment without fitting, and to effect every degree of safety and performance.

4.6.3 Hydrostatic pressure and vacuum test. The hose connection shall be fitted with a blank plug with the exception of a bleed valve installed in the plug. The bleed valve shall be used to purge air from the hose reel assembly piping. The hose reel assembly shall be flooded with fuel and subjected to pressures of 0, 5, 10, 50, 100, 150, and 200 psi and 15 inches Hg vacuum. Each pressure shall be held for 5 minutes. Immediately after each test pressure is reached, the drum assembly shall be rotated a minimum of four times. The hose reel assembly shall be inspected continuously during this test. Any evidence of leakage in or out of the hose reel assembly or other malfunction shall be cause for rejection.

4.6.3.1 Operating hydrostatic pressure test. The hose reel assembly shall be fitted with a hose as required by the specification for the end item equipment. The hose reel assembly shall be subjected to a hydrostatic test pressure of 150 psi for 10 minutes with the hose fully extended. Any evidence of leakage or other malfunction during this test shall be cause for rejection.

4.6.4 Unwind force test. One half of the hose length shall be extended from the hose reel assembly. A spring scale or similar force measuring device shall be attached to the hose near the hose reel, but outside of the hose rollers and other interfering structures. It shall be demonstrated that a force of no more than 15 pounds is sufficient to maintain the velocity of the hose while it is being unwound. In the event there is some difficulty in conducting this test due to the weight of the hose causing either more or less apparent force, the portion of the hose being dragged may be placed on a roller conveyor or similar device. If collapsible hose is being utilized, it shall be evacuated for this test.

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* 4.6.5 Drag brake test. With the test setup as specified in 4.6.4, the drag brake shall be adjusted to provide rewind forces of approximately 20, 35, 50, and 100 pounds. Each of the test force values shall be demonstrated and the exact force in each case recorded.

* 4.6.6 Rewind rate test. With the drag brake adjusted for a 20-pound unwind force (see 3.6.10), the hose shall be fully extended from the reel. The hose shall be rewound on the reel by operating the motor continuously from start to finish. The rewinding operation shall be timed and the average hose rewind velocity determined. The test shall be conducted by allowing the hose to be fully extended on a smooth concrete or similar surface. The nozzle assembly, if attached, may be carried by hand. The power supply voltage shall be recorded during this test. The average hose rewind rate shall be 3 to 3.5 fps.

4.6.7 Rewind force test. With the test setup as specified in 4.6.4, it shall be demonstrated that the motor and rewind mechanism are capable of producing a 175-pound force on the hose without stalling. Sufficient force shall then be applied to open the circuit breaker. This force shall be recorded. The power supply voltage shall be recorded during this test.

* 4.6.8 Structural loading (drum assembly) test. With the drum assembly installed in the hose reel assembly, a 150-pound force shall be applied parallel to the axis of the drum. The load shall be applied in at least four points, one point at a time at the outer periphery of each end disc first in one direction and then the other. The deflection shall be measured at the point the force is applied, while the force is being applied and at any other point subject to substantial deflection. Deflection of more than 3/8 inch at any point shall be cause for rejection.

* 4.6.9 Structural loading (frame) test. With the drum and hub assemblies removed and the frame firmly mounted with the normal mounting hardware, it shall be subjected to a 750-pound load individually at each hub assembly mounting point. The 750-pound load shall be applied vertically downward, parallel to the axis of the drum in first one direction and then the other, and perpendicular to the axis of the drum and parallel to the ground plane in first one direction and then the other. Deflection of more than 3/8 inch at any point shall be cause for rejection.

* 4.6.10 Endurance test. The endurance test shall consist of 1,000 cycles of unwinding and rewinding the hose on the hose reel assembly. At least the first 10 of every 100 cycles shall be consecutive cycles in as rapid succession as possible with the drag brake adjusted to provide a rewind force of 35 pounds as in 4.6.5, to be followed by 90 consecutive cycles with no more than 1 minute between cycles. The average hose unwinding speed shall be 3 to 3.5 fps. During 5 out of every 100 cycles, the hose shall be unwound so

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that the velocity of the hose is 5 fps just before the limit stop position is reached. If appropriate, the limit stop shall be adjusted first at one wrap and then in nine equal increments to zero wrap during the endurance test and satisfactory operation and repeatability of the limit stop shall occur. If collapsible hose is used, it may be handled in the evacuated condition. If noncollapsible hose is used, it shall be handled flooded with fuel. After the 1,000 cycles, the hose reel shall be subjected to the operating hydrostatic pressure test of 4.6.3.1. Any evidence of binding, motor overheating, leakage, or any other malfunction during any portion of this test shall be cause for rejection. After the test has been completed, the hose reel assembly shall be disassembled and all parts inspected. Any evidence of excessive wear, lack of lubrication, overheating, or structural damage to any part or component shall be cause for rejection.

* 4.6.11 Electromagnetic interference test. The hose reel assembly, including all electrical components which could produce electromagnetic radiation, motor, switches, controls, and circuit breaker shall be tested for radiated interference in accordance with MIL-STD-461 (see 6.2).

4.6.12 Environmental tests. The hose reel assembly shall be subjected to the following tests conducted in accordance with the specified methods of MIL-STD-810. The tests of 4.6.2, 4.6.3.1, and 4.6.6 shall constitute the operation to be performed. An engineering evaluation may be performed in lieu of the testing for the tests specified in 4.6.12.3, 4.6.12.4, 4.6.12.5, and 4.6.12.6. The engineering evaluation shall consist of a written detailed analysis of the hose reel assembly and all its parts and components to satisfactorily accomplish the required environmental conditions. (See 6.2.)

4.6.12.1 High temperature test. The hose reel assembly shall be subjected to high temperature testing in accordance with method 501, procedure I.

4.6.12.2 Low temperature test. The hose reel assembly shall be subjected to low temperature testing in accordance with method 502, procedure I. The fuel used to accomplish the testing shall be no warmer than 0°F.

4.6.12.3 Humidity test. The hose reel assembly shall be subjected to humidity testing in accordance with method 507, procedure I.

4.6.12.4 Fungus test. The hose reel assembly shall be subjected to fungus testing in accordance with method 508, procedure I.

4.6.12.5 Salt-fog test. The hose reel assembly shall be subjected to salt-fog testing in accordance with method 509, procedure I.

4.6.12.6 Dust test. The hose reel assembly shall be subjected to sand and dust testing in accordance with method 510, procedure I.

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4.6.13 Reliability demonstration and test. Satisfactory completion of all tests specified herein without failure demonstrates compliance with the quantitative reliability requirements of this specification.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

* 5.1.1 Level A

* 5.1.1.1 Cleaning and drying. The hose reel assembly shall be cleaned in accordance with process C-1 and dried in accordance with procedures D-1 and D-4 of MIL-P-116.

* 5.1.1.2 Preservation application. Exposed metal surfaces shall be preserved with type P-1 preservative of MIL-P-116. Gear and clutch housings shall be filled to operating levels with functional lubricants or greases, as applicable.

* 5.1.1.3 Packaging. The electric motor and control switch shall be removed and packaged in accordance with method IIB of MIL-P-116. The other container shall be a style RSC, class domestic, fiberboard box conforming to PPP-B-636. The box shall be sealed with tape conforming to PPP-T-45, or equal.

* 5.1.2 Level C. Hose reel assemblies shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's commercial practice when it meets the requirements of this level.

* 5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

* 5.2.1 Level A. Hose reel assemblies, preserved and packaged in accordance with 5.1.1, shall be packed one each in an overseas-type, style A or B container conforming to PPP-B-601. The packaged electric motor and control switch shall be secured within the shipping container. Cushioning, blocking, and bracing, as required, shall be in accordance with MIL-STD-1186. The container shall be strapped in accordance with the appendix of the applicable container specification.

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* 5.2.3 Level C. Hose reel assemblies shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's commercial practice when it meets the requirements of this level. Shipping containers shall conform to the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

* 5.3 Marking. In addition to any special markings specified in the contract or order (see 6.2), containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification provides the general requirements for a hose reel assembly intended for use on aircraft fuel servicing vehicles for servicing all types of conventional hydrocarbon fuels.

* 6.2 Detail data for equipment specifications. The following should be noted in the preparation of detail specifications for a specific hose reel and in specifications for end item equipment into which a hose reel will be incorporated:

a. If metals in contact with the fuel should be more restrictive than brass, bronze, or 300 series stainless-steel (see 3.5.2)

b. If product to be handled is other than fuels in accordance with MIL-G-3056, MIL-G-5572, and MIL-T-5624 (see 3.6.4)

c. Maximum flow rate required through the hose reel (see 3.6.4)

d. Specify single-wrap or multi-wrap hose reel (see 3.6.5)

e. Rewind rate if other than 3 to 3.5 fps (see 3.7.3)

f. Type, diameter, and length of hose to be utilized on the hose reel assembly (see 3.8.1.1)

g. If either a pipe thread, flange, or groove-type hose reel inlet connection is required (see 3.8.2.1

h. If inlet hub is to be fabricated of materials other than brass, bronze, or stainless-steel (see 3.8.2.1.)

i. If swing joint should be fabricated of materials other than brass, bronze, or stainless-steel (see 3.8.2.1.1

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j. If interior piping should be fabricated of materials other than brass, bronze, or stainless-steel (see 3.8.2.3)

k. If hose reel hose connection should be other than a male thread (see 3.8.2.3.1)

l. If hose reel motor and the electrical components should be other than 12-V dc (see 3.8.3.1)

m. If circuit breaker should be installed other than external to the hazardous vapor area (see 3.8.3.4.2)

n. If a dual reel frame is not allowable where applicable (see 3.8.4.2)

o. If finish paint should be other than gloss yellow, code DG (see 3.12.2)

p. If test fluid should be other than grade JP-4 turbine fuel conforming to MIL-T-5624 (see 4.3.2)

q. Location and conditions for preproduction testing (see 4.4)

r. If electromagnetic interference tests are not required on the hose reel assembly since they will be conducted on the complete end item equipment on which the hose reel is incorporated (see 4.6.11)

s. If actual testing is required in lieu of an engineering evaluation for humidity, fungus, salt-fog, and dust (see 4.6.12)

t. When hose reels are being provided installed in end item equipments, the following tests should be conducted on the hose reel after installation on the end item both as a part of the preproduction tests, and individual acceptance tests on the end item equipments. These tests should be specified in the specification for the end item equipment:

(1) Mechanical operation

(2) Unwind force

(3) Rewind rate

(4) Rewind force

u. Required levels of preservation and packaging, and packing (see 5.1 and 5.2)

v. Special shipment marking (see 5.3).

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6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability that an item will perform a specified function without failure under given conditions for a specified period. The minimum reliability is calculated from the demonstrated minimum MTBF and the mission time as follows:

$$R_{min.} = e^{-t/MTBF}$$

6.3.2 Mean-time-between-failure. The mean-time-between-failure (MTBF) is defined as the arithmetical mean (average) of the operating time between failures. A lower confidence limit or minimum MTBF can be obtained from demonstration data (see 4.6.13).

6.3.3 Mission time. Mission time (t) is defined as the expected or average operating time for each use of an item and for this specification is defined as 0.25 hour.

* 6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

dc	direct current
fps	feet per second
Hg	mercury
ID	inside diameter
MTBF	mean time between failure
psi	pounds per square inch
V	volt

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* 6.5 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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